A SNOW-BASED CLOSURE PHASE MODEL FITTED TO THE SEASONAL TREND FROM 2017 TO 2000 OVER GREENLAND

Zezhong Wang[1], Paco López-Dekker[1], and Lorenzo Iannini[1]

[1] Geoscience and Remote Sensing, Delft University of Technology, 2628 CN Delft, The Netherlands

ABSTRACT

Phase inconsistency, a phenomenon found in interferometric SAR, quantified by triplet closure phase, is mathematically resulted from average process, containing the volume scattering information. In this paper, we proposed a snow-based exponential decay model to explain the seasonal trends of the space texture of Sentine-1 closure phases from 2017 to This model builds direct relationship between the statistical characteristics of closure phase and penetration depth of radar in snow.

***Keywords*—** Phase inconsistency, interferometric SAR, snow, Greenland

1. INTRODUCTION

Nonzero closure phase, or phase inconsistency, can be observed in a triangle on curved surface, and that also happen on radar signals discovered by De Zan et al. (2015) [1]. The sum of spatially averaged interferograms of SAR in a circular way might or might not be close to zero. Volume scattering could be used to model the phase inconsistency. De Zan et al. characterized the volume scattering with a real-valued vertical scattering profile function and found that the excess of closure phase is directly dependent on the profile skewness at the condition of small baselines [2]. However, the relationship of coherence and closure phase is not directly interpreted in the model. Yet coherence makes a very difference on the closure phase.

In this paper, we study further insight into phase inconsistency and volume scattering with case of Sentinel-1 observation on Greenland. An interesting finding is that, due to the temperature change, the space-texture of closure phase shows an annually trend from 2017 to 2020. According to this funding, we built an exponential decay model to explain it by relating the closure phase and coherence of SAR to penetration depth of snow. The snow-based model proposed in this paper is different with the soil-based model for the expectation of closure phase is skewness independent so that only the penetration depth left unknown directly responding to the statistical characteristics of closure phase.

2. PHASE INCONSISTENCE AND SOIL-BASED MODEL

Phase consistency is defined as the sum of the interferometric phase of three SAR images in circle order:

$$\Phi_{123}(0) = \phi_{12} + \phi_{23} + \phi_{31} \tag{1}$$

The sum of the averaged interferometric phase of three SAR images in circle order phase might or might not be zero. The nonzero phenomenon is called phase inconsistence, and phase excessing zero is defined as closure phase:

$$\Phi_{123} = \bar{\phi}_{12} + \bar{\phi}_{23} + \bar{\phi}_{31} \tag{2}$$

De Zan et al. have related the closure phase to the real-valued vertical scattering profile in terms of wavenumber (k_z) with the height coordinate (z):

$$\begin{aligned}\Phi_{123} &= \bar{\phi}_{12} + \bar{\phi}_{23} + \bar{\phi}_{31} \\ &\approx \phi'(0)(k_{12} + k_{23} + k_{31}) \\ &\quad + \phi'''(0)(k_{12}^3 + k_{23}^3 + k_{31}^3)\end{aligned} \tag{3}$$

where ϕ is the phase component of the interferogram in polar form as:

$$I(k_z) = A(k_z)e^{-j\phi(k_z)} \tag{4}$$

In this way, both the amplitude and phase of the interferogram are functions of k_z and giving the trivial consistency of wavenumbers ($k_{12} + k_{23} + k_{31} = 0$) in (3), the closure phase becomes a function of profile skewness:

$$\Phi_{123} = \frac{1}{2}E[(z - E(z))^3]k_{12}k_{23}k_{31} \tag{5}$$

In terms of SAR images, results of radar backscattering, Equation (5) means that the phenomenon of phase inconsistence or the value of closure phase responds to the wavenumber of scattering profile. Then, the volume scattering model also responds to the wavenumber of scattering profile will be introduced in next subsection.

3. SNOW-BASED MODEL

In terms of the glacial snow, almost everything is dynamic to observers. At least, the position of ground keeps changing due to the motion of ice-shelf; and the thickness of snow varies corresponding to the weather and temperature. With a temporal decorrelation with time-lag τ for the time delay of repeat-pass SAR interferometric [3], a linear vertical profile function $v_r(z)$ of line-of-sight velocities for the motion of

snow layers, and the penetration depth δ_Z for thickness, the volume scattering in snow can be modeled as:

$$\eta_d(z) = \gamma_t(\tau)\frac{\sigma_o}{\delta_z}u(z_o - z)e^{(z-z_o)/\delta_z}e^{-j2k_o v_r(z_o)\tau}e^{-j2k_o v_r'(z-z_o)\tau} \quad (6)$$

where the density of snow is assumed as exponential decay, z is the height coordinate, σ_0 and z_0 are the normalized radar cross section and the height of the surface above the reference height, respectively. u(·) is the unit step function. Thus, the interferogram of a pair of SAR acquisitions is associated with a wavenumber (k_z) as

$$I(k_z) = \int_{-\infty}^{\infty} \eta_d(z)e^{-jk_z z}dz \quad (7)$$

Then, the casualty of phase inconsistence in SAR image and volume scattering in snow integrating equating (4) and (7), and since the skewness of the exponential distribution is constant of 2, the equation (5) become more concise

$$\Phi_{123} = k_{12}k_{23}k_{31} \quad (8)$$

Thus, in the model, the only involved variance is the wavenumber building a bridge between SAR image and scatterers. In next subsection, we will discuss about how the wavenumber transfers the information from nature phenomenon to closure phase.

4. THE RELATIONSHIP AMONG CLOSURE PHASE, COHERENCE AND PENETRATION DEPTH

For the equation (7) we discussed before, only two parameters, the motion of ice-sheet and penetration depth, impacting the cross-correlation of normalized interferogram and profile as the normalized integral result:

$$\gamma(k_z) = \frac{1}{1 - j(k_m + \delta_z \cdot k_z)}e^{-j[(\phi_o + \phi(k_z)]} \quad (9)$$

It is the same form of the interferometric coherence where both k_m and ϕ_o respond to the motion, and when they are zero with assumption of ice-sheet keeping still for the repeat-pass SAR acquisitions, $\gamma(k_z)$ has Hermitian symmetry with respect to k_z. As shown in Fig. 1, the penetration depth decides the shape of cross-spectral. And the velocity of ice-sheet motion decides the peak position of the cross-spectra. While wavenumber closes to zero, the increase of penetration depth will lead to an increase of slope. Concerning the inevitable bias of estimation, the coherence obtained from real SAR image is not an exact value but more likely a range of coherences with a confidence interval. Thus, as shown in Fig. 2a, the variation of wavenumber in the coherence interval of [0.2, 0.4] is significantly higher than that of [0.7, 0.9], and moreover, the slope of wavenumber is also increase with penetration depth.

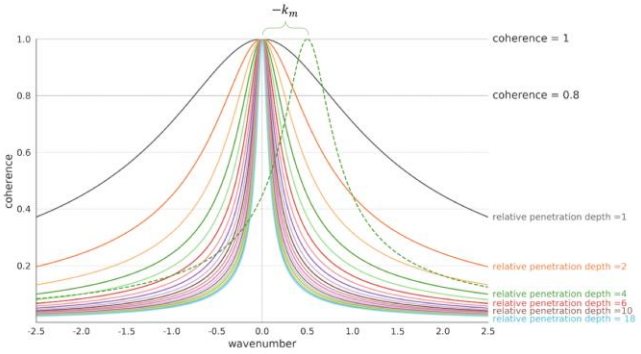


Fig. 1. The cross-correlation of coherence and wavenumber

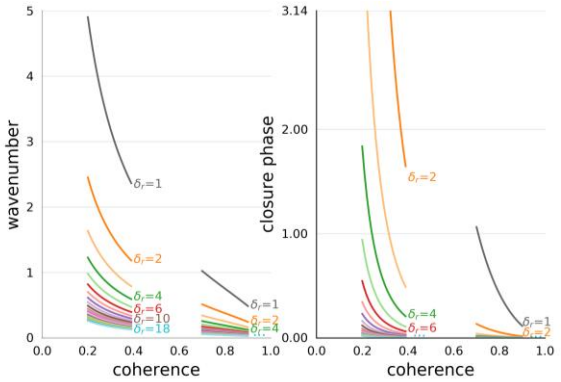


Fig. 2. The variation of wavenumber and closure phase for different coherence interval

In a time-series observation of SAR, the sequential changes of spatial autocorrelation of closure phase have potential to indicate a seasonal trend of radar penetration depth. The change of penetration depth is probably a synthetic result of snowfall, melting, and refreezing. In next section, we will show how to find a physically phase closure in SAR images by observation in Greenland.

5. RESULTS

Glacier should show some seasonal characteristics along with the temperature and precipitation. By using more than 200 scenes of Sentinel-1 C-band SAR image from 4th January 2017 to 09th October 2020, we obtained 166 sequential closure phase samples in one site located in the southwest of Greenland (Fig. 3).

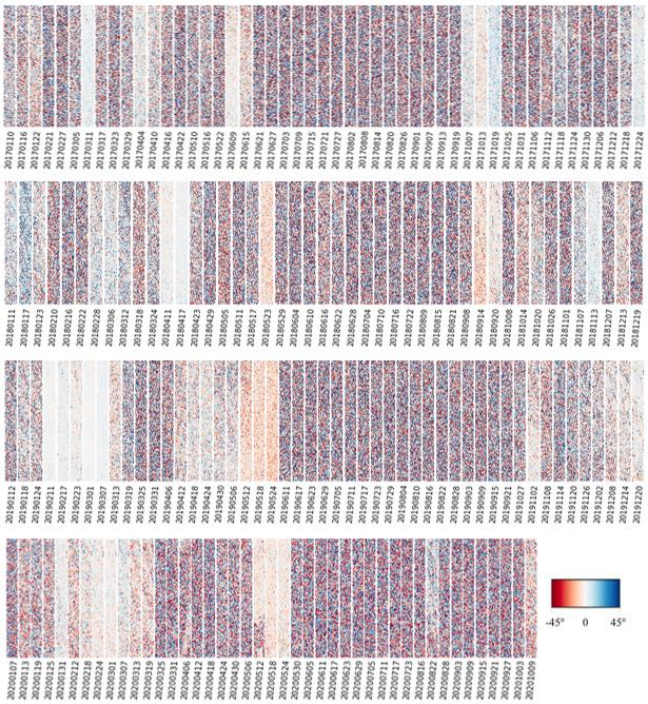


Fig. 3. The closure phase from time-series observations of SAR from 2017 to 2020 over one site in the southwest of Greenland.

It can be found that the spatially stable closure phase only appeared in the colder season but never happened in the summer from 2017 to 2020. Given focusing on the summer season, it is interesting that the period from disappear of spatially stable closure phase to its reappearance, e.g., that time interval is 90 days for 2017 (from 21-06-2017 to19-09-2017), 100 days for 2018 (from 29-05-2018 to 08-09-2018), but more than 120 days for 2019 and 2020 (from 11-06-2019 to 27-10-2019; from 30-05-2020 to 03-10-2020). However, it seems that the spatially stable closure phase just randomly appeared in the other seasons.

To quantitively statistic the seasonal and annual trends of the closure phase, we introduced *Moran's I*, a correlation coefficient that measures the overall spatial autocorrelation of data set, to mark each set of closure phase a value from -1 to 1, where a perfect clustering of similar values will result in 1 and a perfect randomness or no spatial autocorrelation will be marked as 0, but a perfect clustering of dissimilar values will lead to -1. As shown in Fig. 4, from January 2017 to May 2020, *Moran's I* presents two significant peaks at 17th April 2018 and 24th May 2019 for the 166 closure phase samples. Selecting some peaks of *Moran's I*, we have drawn a periodic envelope (the dashed line in Fig. 4)

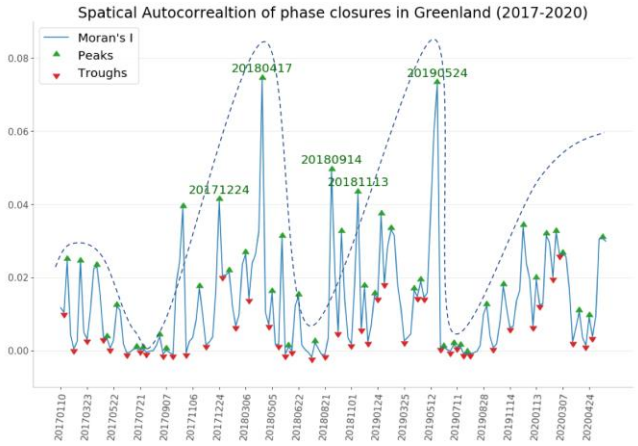


Fig. 4. The Moran' I variation of closure phase from 2017 to 2020 over one site in the southwest of Greenland.

which presents troughs at July and peaks at about two months before July, and the uptrend costs more time (about 7 month) than the downtrend (about 2 month). However, there are also some values of *Moran's I* do not fit to this envelope, e.g., 20180914 and 20181113. Moreover, the findings in closure phase could be concluded in three points: 1) annual changes of the disappearance period; 2) seasonally periodic variation; and 3) the months in summer performing lowest value of *Moran's I*. Those phenomena found in closure phase may be related to the snowfall, melting, and refreezing responding to the penetration depth of microwave.

The relative permittivity of water and glacial ice (80 ~ 120) is much higher than that of dry snow (4 ~ 30) [4] and the penetration depth is major depend on the thickness, density and water content of snow upon ice. In the summer season, when the temperature is over 0℃, the snow is melting resulting in an increase of the permittivity and a decrease of penetration depth. The annual disappearance of closure phase probably suggests that the melting process of snow probably starts from May and lasts about two months, and that the accumulation of snow will re-start in September or October when the day temperature decline below 0℃. The extension of this period from 2017 to 2019 might be account for the global warming.

Providing that the snowy weather happens uniformly over one year but the amount of each snowfall is random, the penetration depth will depend on the both of soft new snow (with permittivity of 5) and accumulated old snow. Considering the accumulation

process, the penetration depth will roughly increase as the rise part of envelope in Fig. 4.

In addition, a heavy snowfall contributing a thick layer of soft new snow will make a big difference on the radar penetration depth then even impact the closure phase, but when the snowfall happen is almost random. Also, the soft new snow in winter might contribute to a higher coherence as shown in Fig. 5.

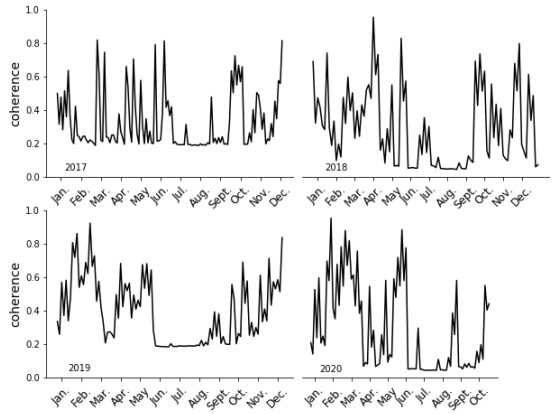


Fig. 5. The average coherence of time-series interferometric SAR from 2017 to 2020 over one site in the southwest of Greenland

The change of penetration depth is resulted from snowfall, compression of snow layer and melting. Then, we also have modeled this phenomenon. The physical thickness of snow covering on the ice surface will rise after precipitation, but later the compression and melting will happen. We used a discrete convolution model to describe this process:

$$S(n) = f[k] * h[n-k] = \sum_{k=-\infty}^{k=\infty} f[k]u[n-k]a^{[n-k]} \qquad (10)$$

where $f[n]$ is a unit step function, n and k represent the time and time delay. The relative permittivity of soft new snow with density 0.13 g./cm.^{3} is 4, and it will rise to a higher level with increasing density. The higher relative permittivity is, the lower penetration depth become. Then, the simulated penetration depth is shown in Fig. 6, and formulated as:

$$p(n) = f[k] * h'[n-k] = \sum_{k=-\infty}^{k=\infty} f[k]u[n-k]b^{[n-k]} \qquad (11)$$

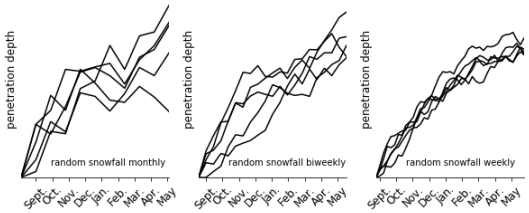


Fig. 6. The simulation of penetration depth with random snowfall monthly, biweekly and weekly by using a convolution model

The three graphs in Fig. 6 are corresponding to different sample rate, where the small sample rate (weekly) results in a clearer trend than the ones with larger sample rates (biweekly and monthly).

Comparing Fig. 6 with Fig. 4, the simulation trends of penetration depth have similar shape with the Moran' I curve of closure phase.

6. CONCLUSIONS

Phase inconsistence phenomenon is also found in Greenland and it shows a seasonal trend from 2017 to 2020, and it is possible resulted from the precipitation of snow and melting. Considering the dynamic process of precipitation, we proposed a snow-based closure phase model to fit the trend. In the model, the statistical characteristic of closure phase responds to penetration depth, which can be used to interpret that the spatial autocorrelation of phase closure is low in summer but higher in winter.

7. REFERENCES

[1] F. De Zan, M. Zonno, and P. Lopez-Dekker, "Phase inconsistencies and multiple scattering in SAR interferometry," *IEEE Trans. Geosci. Remote Sen.*, vol. 53, no. 12, pp. 6608-6616, 2015

[2] F. De Zan and G. Gomba, "Vegetation and soil moisture inversion from SAR closure phases: First experiments and results," *Remote Sens. Env.*, vol. 217, pp. 562-572, 2018.

[3] P. López-Dekker, F. De Zan, S. Wollstadt, and et al., "On the tomographic information in single pairs of crossing-orbits SAR acquisitions" Proceedings of IEEE IGARSS 2011, Vancouver, BC, Canada.

[4] S. Evans, "Dielectric properties of ice and snow–a review", *J. Glaciol.*, vol. 5, no. 42, pp. 773-792, 1965.